

SHIVAM JOSHI

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Full-Stack Developer with 3 self-initiated, portfolio-grade projects spanning AI/voice pipelines, real-time WebRTC systems, payments, and full CRUD platforms. Skilled across MERN, TypeScript, and PostgreSQL/Prisma, with end-to-end ownership from UI to backend APIs to deployment.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, HTML5, CSS3, SQL

Frontend: React.js, React Router, Vite, Tailwind CSS, Bootstrap, Context API, Axios, MediaRecorder/Browser Media APIs

Backend: Node.js, Express.js, Bun, REST APIs, FastAPI, JWT, bcrypt, Middleware, CORS

Real-Time / Communication: Socket.IO, WebRTC, SDP, ICE candidates, RTCPeerConnection, peer-to-peer signaling

Databases: MongoDB, MongoDB Atlas, Mongoose, PostgreSQL, Prisma, Redis

AI / Voice: Google Gemini API, LLM integration, RAG (Retrieval-Augmented Generation), Whisper/faster-whisper, Kokoro TTS, prompt/context engineering

Payments / External Services: Stripe, Stripe Webhooks, Cloudinary, OMDb API, EmailValidation.io, GitHub API

Tools / DevOps: Git, GitHub, Docker, Docker Compose, Turborepo, Postman, VS Code, Vercel

PROJECTS

AI-Powered Voice Interviewer | [Live](#) | [GitHub](#)

- Built a full-stack AI mock-interview platform with a 6-stage interview workflow (Introduction, Resume, GitHub, Technical, HR, Feedback) using React, TypeScript, Node/Express, and Prisma/PostgreSQL.
- Engineered a real-time voice pipeline (MediaRecorder → Whisper/faster-whisper speech-to-text → LLM → Kokoro TTS) for spoken AI-generated interview questions.
- Integrated resume PDF parsing and GitHub data to generate candidate-specific, context-aware questions with full conversation history tracking.
- Delivered AI-generated post-interview analytics scoring technical performance, communication, and confidence, deployed as Dockerized microservices in a Bun/Turborepo monorepo.
- Implemented a RAG (retrieval-augmented generation) system to ground AI-generated interview questions and feedback in relevant candidate/resume context.
- Key takeaway: Orchestrating a multi-stage async pipeline (MediaRecorder → speech-to-text → LLM → TTS) across separate Dockerized services meant a single dropped audio chunk or slow LLM response could cascade through the whole chain — this pushed toward designing for graceful failure and explicit state reconciliation across services, not a linear, always-succeeding request flow.

Veyro — Real-Time Communication & Video Meeting Platform | [Live](#) | [GitHub](#)

- Developed a full-stack messaging and video platform (React, TypeScript, Node/Express, MongoDB, Socket.IO) supporting 4 real-time communication modes: 1-on-1 chat, audio calls, video calls, and multi-participant meetings.
- Implemented WebRTC peer-to-peer audio/video with Socket.IO signaling, SDP offer/answer exchange, ICE candidate handling, and screen sharing via track replacement.
- Designed meeting infrastructure with host controls (mute/remove participants, end-for-everyone), unique meeting codes, and race-condition-safe signaling validation.
- Secured the platform with JWT authentication, bcrypt password hashing, and Cloudinary media management.
- Key takeaway: In mesh WebRTC — where each participant holds a direct peer connection to every other participant — connection count grows combinatorially and race conditions surface easily when peers join or leave mid-signaling; this meant treating signaling as an explicit state machine with valid transitions, not a sequence of one-off events, and handling renegotiation and stale-connection cleanup deliberately.

QuickBasket — Full-Stack Grocery Delivery Platform | [Live](#) | [GitHub](#)

- Developed a full-stack grocery delivery platform (React, Node/Express, MongoDB) with product discovery, search, category filtering, and cart management.
- Implemented 2 checkout flows — Cash on Delivery and Stripe payments with webhook handling — alongside JWT-secured customer and seller authentication.
- Architected a seller dashboard for product and order management plus customer order history, deployed on Vercel with MongoDB Atlas and Cloudinary.
- Key takeaway: Payment webhooks can arrive out of order or be delivered more than once, so handling them idempotently — ensuring a duplicate Stripe event never double-processes an order — showed that correctness in a distributed system is less about the happy path and more about designing for duplicate, delayed, or out-of-order events.

EDUCATION

VIT Bhopal University — B.Tech, Computer Science & Engineering, 2026

ACHIEVEMENTS

- Solved 200+ LeetCode problems in Java, covering arrays, strings, recursion, trees, BFS/DFS, sliding window, and dynamic programming.

ADDITIONAL PROJECTS

Movie Explorer (vanilla JS + OMDb API) · Email Validator (REST API integration) · Spotify Clone (HTML/CSS/JS) · Task Manager App · Calculator · Number Guessing Game · 3D Interactive Portfolio (React)